

COMPARISON BETWEEN INTRINSIC AND EXTRINSIC OPTICAL-FIBRE SENSORS

Intrinsic-type optical-fibre sensors	Extrinsic-type optical-fibre sensors
Application: Rotation, acceleration, strain, pressure and vibration	Applications: Temperature, pressure, liquid level and flow
More sensitive	Less sensitive
Tougher to multiplex	Easily multiplexed
Reduces connection problems	Connection problems are more
Tougher to use	Easier to use
More expensive	Less expensive

eter, which is located outside the engine. In the same way, these sensors are used to measure internal temperature of transformers.

A comparison between intrinsic and extrinsic optical-fibre sensors is given in the table.

Advantages are many

Optical-fibre sensors deliver the following advantages:

- Contain no moving parts or electrical circuitry; hence, are immune to all forms of electrical interferences
- No possibility of spark, so are safe to be used in hazardous sensing environments, such as oil refineries, grain bins, mining operations, pharmaceutical manufacturing and chemical processing
- No danger of electrical shocks while repairing broken fibres
- Small and lightweight
- Have excellent range, resolution and multiplexing capabilities
- Resistant to high temperature and explosive environment

Major disadvantages that need solutions

- Complex detection system
- Expensive
- Non-familiarity to end user
- Requirement of precise installation procedures
- Usable measuring system's development is complex

Fibre-optic sensors gaining ground

It is believed that fibre-optic sensors are the best way to drive a nation's sensor technologies. However, one must ensure that the use of these sensors is profitable at the same time. These sensors find vast applications not only in industries but also in sectors like healthcare, aerospace and marine.

Hydrophones are used for seismic and sonar applications. Hydrophones with more than 100 sensors per fibre cable have been developed. Hydrophone sensor systems are used by oil industries and in the navies of some countries.

During MRI-guided surgery, a fibre-optic microphone is used in the MRI machine.

Typical fibre-optic sensor applications can be summarised as follows:

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